

Dwell-Conditioned Cycle-to-Cycle Switching-Time Feedback for Mode-Dependent Switched-Affine Systems

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Abstract

Static cycle-to-cycle feedback regulates a reference periodic trajectory by shifting the interior switching instants of a fixed-period mode sequence. Its raw timing action can nevertheless produce inadmissibly short or negative dwell durations. We derive the one-cycle model for mode-dependent switched-affine dynamics and condition the complete raw action by its largest dwell-feasible scalar factor. The conditioner preserves the action direction, mode order, and cycle period and requires an $\mathcal{O}(N)$ scan without an online solver. Uniform scaling places the conditioned linearized matrices on a segment between the nominal cycle matrix and the raw closed-loop matrix. An offline semidefinite program seeks a common quadratic Lyapunov function at these two endpoints. We prove that strict endpoint decrease implies uniform exponential stability for every sequence of conditioning factors in $[0, 1]$. A three-cell DC–DC converter benchmark admits this certificate. Exact switched-affine simulations compare the controlled and open-loop three-dimensional trajectories and show eventual open-loop convergence to the same reference trajectory. Conditioned feedback reaches a normalized error below **0.01** at cycle 6, compared with persistent open-loop crossing at cycle 22009, while enforcing the assumed 3 μ s dwell bound. The common Lyapunov result concerns the linearized family; exact nonlinear stability remains local, and the large-error response is simulation evidence.

Keywords: switched-affine systems, periodic trajectories, switching-time control, dwell-time conditioning, common quadratic Lyapunov function, semidefinite programming

1 Introduction

Power converters control voltage and current by moving through a sequence of switch configurations. In practice, the switching pattern must respect electrical limits, switching frequency, a minimum dwell duration for each configuration, and a short computation time [Karamanakos et al. \(2020\)](#). Keeping the individual configurations explicit gives a hybrid-affine or switched-affine model [Albea-Sanchez et al. \(2021\)](#). Repeating the pattern produces a periodic trajectory rather than a

constant state, and controlling such trajectories is an established problem for switched-affine systems [Egidio et al. \(2020\)](#); [Xu and Lazar \(2024\)](#). We consider regulation at a selected cycle anchor. At each cycle start, the controller steers the state toward that point on the reference periodic trajectory.

Predictive control tackles the problem by solving a constrained optimization problem online. Recent work reduces the online burden through Laguerre parameterization, dedicated fixed-frequency solvers, and candidate pruning [Saeed et al. \(2022\)](#); [Yang et al. \(2022\)](#); [Augustine and Patil \(2023\)](#).

Controllers also differ in what they may change. State-dependent switching laws choose the active subsystem [Egidio et al. \(2022\)](#), where switching-time control keeps an ordered sequence and moves its transition instants. We take this same approach, changing the transition instants and keeping the rest of the parameters the same. At each cycle start, the error is computed and the offsets applied during that cycle, with no online solver.

Patino et al. studied mode-dependent cyclic systems and how each switching instant affects the trajectory. They applied an online predictive controller to adjust the switching instants and track an optimal periodic trajectory [Patino et al. \(2010\)](#). Marcolino et al. considered a fixed-period cycle controlled by changing its interior switching instants. They used a discrete LQR and dwell-constrained predictive control to apply corrections based on the one-cycle model [Marcolino et al. \(2021\)](#). Their model has a common state matrix with switched actuator levels. Each interior switching instant separates two dwell durations, corresponding to the modes active before and after it. Delaying the instant lengthens the preceding dwell and shortens the following one, so the dwell limits are coupled. The predictive controller enforces these limits through online optimization, and its feasible set is characterized through polyhedral projection [Marcolino et al. \(2021\)](#). The standalone LQR comparison instead clips the offsets one at a time, which changes the requested direction and may shift the final scheduled instant [Marcolino et al. \(2021\)](#). Projecting a raw action directly onto the feasible set would require a quadratic program and would generally change its direction. Direction-preserving control allocation scales the complete command, but it also uses online optimization and does not address fixed-period dwell constraints [Nakamura-Zimmerer and Miller \(2026\)](#). The remaining problem is to find the largest feasible scale directly, preserving the timing-action direction, mode order, and cycle period without an online solver.

We adapt direction-preserving scaling to fixed-period switching schedules. Each dwell constraint gives an upper bound on the scale, and the smallest bound is the largest feasible factor in $[0, 1]$. Checking all N bounds takes $O(N)$ time and requires no online solver. Scaling the complete raw offset vector preserves its direction, the mode order, and the cycle period. For the linearized raw

closed loop, an offline polyhedral iteration finds the maximal invariant subset of the Raw-Action Admissible Region. Outside this set, the conditioner guarantees current-cycle dwell feasibility but not convergence; the nonlinear guarantee remains local. [Section 2](#) derives the cycle model, [section 3](#) develops the conditioner and invariant region, and [section 5](#) presents the converter study.

2 Problem formulation and one-cycle model

The control objective is to approach a prescribed periodic trajectory without changing its mode sequence or cycle period. We first express this objective at cycle starts, then derive how shifts of the interior switching instants change the next cycle-start state.

2.1 Plant, reference trajectory, and timing coordinates

Consider the switched-affine plant

$$\dot{\mathbf{x}}(t) = A_{\sigma(t)}\mathbf{x}(t) + \mathbf{b}_{\sigma(t)}, \quad \mathbf{x}(t) \in \mathbb{R}^n. \quad (1)$$

Every cycle visits the fixed sequence $(q_1, \dots, q_N)$. Its nominal cycle-relative boundaries and dwell durations are

$$\begin{aligned} 0 = \bar{t}_0 < \bar{t}_1 < \dots < \bar{t}_N = T, \\ \bar{d}_i = \bar{t}_i - \bar{t}_{i-1}. \end{aligned} \quad (2)$$

Repeating this schedule generates the reference periodic trajectory $\bar{\mathbf{x}}(t)$. Its cycle anchor is $\mathbf{x}^* = \bar{\mathbf{x}}(0) = \bar{\mathbf{x}}(T)$. With $\mathbf{x}_k = \mathbf{x}(kT)$, the cycle-start error is $\mathbf{e}_k = \mathbf{x}_k - \mathbf{x}^*$. The target is this phase-aligned trajectory, not a constant average operating point.

The controller moves only the $N - 1$ interior instants:

$$\begin{aligned} t_{k,i} &= \bar{t}_i + \delta\tau_{k,i}, \quad i = 1, \dots, N - 1, \\ \delta\tau_{k,0} &= \delta\tau_{k,N} = 0. \end{aligned} \quad (3)$$

During interval i of cycle k , the plant uses mode q_i between the absolute times $kT + t_{k,i-1}$ and

$kT + t_{k,i}$. The applied dwell vector satisfies

$$\begin{aligned} \mathbf{d}_k &= \bar{\mathbf{d}} + \delta \mathbf{d}_k, & \delta \mathbf{d}_k &= D \delta \boldsymbol{\tau}_k, \\ D_{ij} &= \begin{cases} 1, & i = j, \\ -1, & i = j + 1, \\ 0, & \text{otherwise,} \end{cases} \end{aligned} \quad (4)$$

where $D \in \mathbb{R}^{N \times (N-1)}$. Thus delaying one instant lengthens the preceding interval and shortens the following one by the same amount. Since $\mathbf{1}^\top D = 0$, every offset vector preserves T . The remaining requirement is $\mathbf{d}_k \geq d_{\min} \mathbf{1}$ for a specified $d_{\min} > 0$.

Figure 1(a)–(b) illustrates the timing coordinates using the converter schedule introduced in section 5. In panel (b), delaying instant 8 by 12 μs lengthens dwell 8 and shortens dwell 9 by the same amount, with all other offsets zero. The shortened dwell remains above the assumed 3 μs bound. This isolated offset illustrates the dwell coupling; panel (c) instead shows the aggressive controller's first-cycle raw request.

2.2 Exact one-cycle map

Introduce the augmented state, interval generator, and nominal transition matrix

$$\begin{aligned} \chi &= \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}, & F_i &= \begin{bmatrix} A_{q_i} & \mathbf{b}_{q_i} \\ \mathbf{0}^\top & 0 \end{bmatrix}, \\ E_i &= e^{F_i \bar{d}_i}. \end{aligned} \quad (5)$$

On interval i , $\dot{\chi} = F_i \chi$, so the exact cycle propagation is

$$\chi_{k+1} = e^{F_N d_{k,N}} \dots e^{F_1 d_{k,1}} \chi_k. \quad (6)$$

The rightmost factor acts first; matrices belonging to different modes cannot generally be interchanged. Define

$$\Pi = [I_n \ \mathbf{0}], \quad \iota = \begin{bmatrix} I_n \\ \mathbf{0}^\top \end{bmatrix}. \quad (7)$$

The physical one-cycle map is

$$\begin{aligned} \mathcal{P}(\mathbf{x}, \delta \boldsymbol{\tau}) &= \Pi e^{F_N [\bar{d}_N + (D \delta \boldsymbol{\tau})_N]} \dots \\ &\quad e^{F_1 [\bar{d}_1 + (D \delta \boldsymbol{\tau})_1]} \begin{bmatrix} \mathbf{x} \\ 1 \end{bmatrix}. \end{aligned} \quad (8)$$

The reference schedule and anchor satisfy $\mathcal{P}(\mathbf{x}^*, \mathbf{0}) = \mathbf{x}^*$. They are given design data, not optimization variables of the timing controller.

2.3 Linearization in the independent switching instants

For a dwell change δd_i , the single-interval expansion is

$$e^{F_i (\bar{d}_i + \delta d_i)} = E_i + F_i E_i \delta d_i + \mathcal{O}(\delta d_i^2). \quad (9)$$

Expanding the ordered product, while retaining only terms containing one dwell change, gives

$$\begin{aligned} &e^{F_N (\bar{d}_N + \delta d_N)} \dots e^{F_1 (\bar{d}_1 + \delta d_1)} \\ &= \tilde{\Phi} + \sum_{i=1}^N J_i \delta d_i + \mathcal{O}(\|\delta \mathbf{d}\|^2), \\ &\tilde{\Phi} = E_N \dots E_1, \\ &J_i = E_N \dots E_{i+1} F_i E_i \dots E_1. \end{aligned} \quad (10)$$

An empty product is the identity. In particular, $J_1 = E_N \dots E_2 F_1 E_1$ and $J_N = F_N E_N \dots E_1$. Write $\chi_k = \chi^* + \iota \mathbf{e}_k$, with $\chi^* = [(\mathbf{x}^*)^\top \ 1]^\top$. Since $\tilde{\Phi} \chi^* = \chi^*$, subtracting the nominal propagation cancels the zeroth-order term. The products $\delta d_i \mathbf{e}_k$ and $\delta d_i \delta d_j$ are second order, leaving

$$\begin{aligned} \chi_{k+1} - \chi^* &= \tilde{\Phi} \iota \mathbf{e}_k + \sum_{i=1}^N J_i \chi^* \delta d_{k,i} \\ &\quad + \mathcal{O}((\|\mathbf{e}_k\| + \|\delta \mathbf{d}_k\|)^2). \end{aligned} \quad (11)$$

Projecting onto the physical state and using $\delta \mathbf{d}_k = D \delta \boldsymbol{\tau}_k$ yields

$$\begin{aligned} \mathbf{e}_{k+1} &= \Phi \mathbf{e}_k + \Gamma_\tau \delta \boldsymbol{\tau}_k + \mathcal{O}((\|\mathbf{e}_k\| + \|\delta \boldsymbol{\tau}_k\|)^2), \\ \Phi &= \Pi \tilde{\Phi} \iota = e^{A_{q_N} \bar{d}_N} \dots e^{A_{q_1} \bar{d}_1}, \\ G_d &= [\Pi J_1 \chi^* \ \dots \ \Pi J_N \chi^*], \\ \Gamma_\tau &= G_d D. \end{aligned} \quad (12)$$

The remainder is understood in fixed state and timing units. The augmented product has the block form

$$\tilde{\Phi} = \begin{bmatrix} \Phi & \mathbf{h} \\ \mathbf{0}^\top & 1 \end{bmatrix}. \quad (13)$$

Its unit eigenvalue belongs to the auxiliary constant coordinate. Stability is tested on the physical error dynamics, not on this augmented matrix.

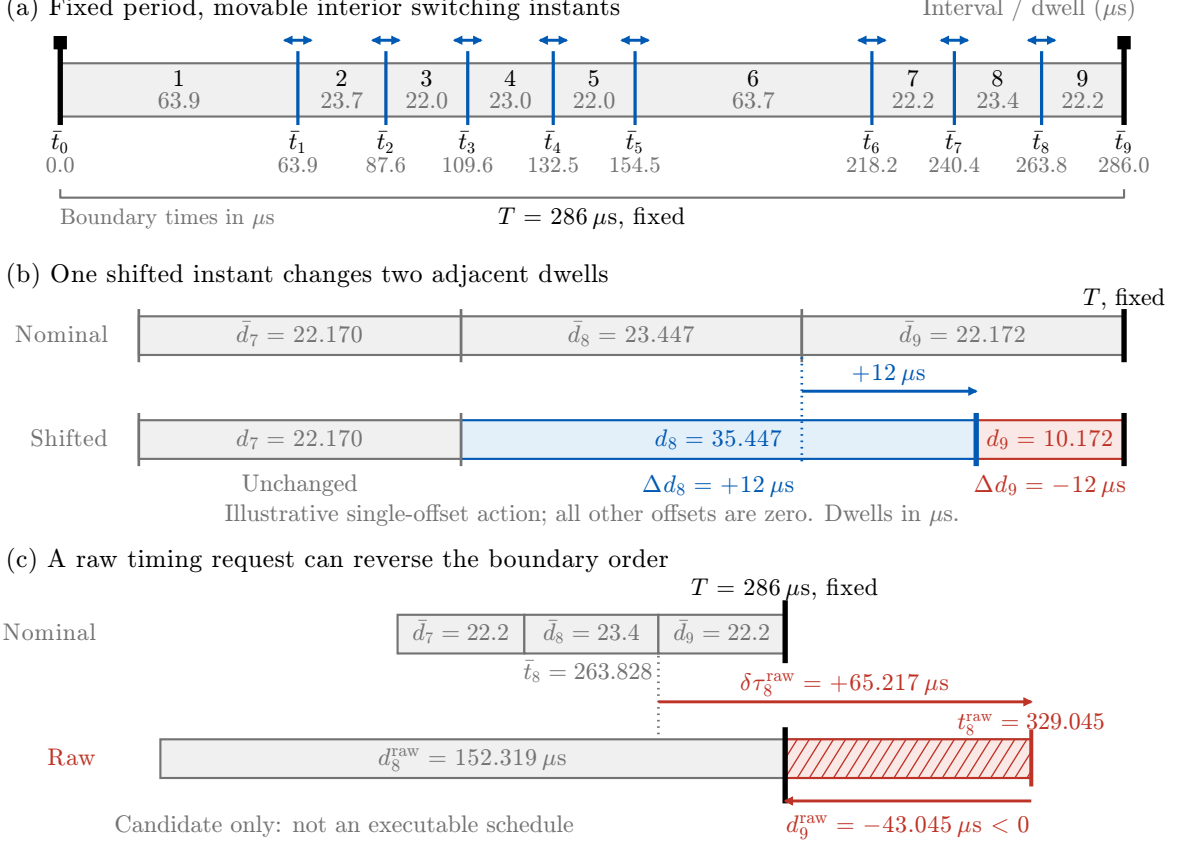


Figure 1 Switching-instant adjustments in the nine-interval schedule. (a) Nominal schedule with eight adjustable interior switching instants. (b) Illustrative $+12 \mu s$ offset of instant 8. (c) Inadmissible first-cycle raw request from the aggressive controller. Hatching denotes a negative candidate dwell.

Each column of D subtracts two adjacent dwell sensitivities. Factoring them gives

$$\Gamma_{\tau,i} = e^{A_{q_N} \bar{d}_N} \dots e^{A_{q_{i+1}} \bar{d}_{i+1}} \times [(A_{q_i} - A_{q_{i+1}}) \bar{x}(\bar{t}_i) + \mathbf{b}_{q_i} - \mathbf{b}_{q_{i+1}}] \quad (14)$$

The bracket is the change in the affine vector field caused by trading time between adjacent modes. The remaining transitions carry this effect to the cycle end. This recovers the switching-sensitivity structure in [Patino et al. \(2010\)](#); switching-time sensitivities also appear in the hybrid limit-cycle analysis of [Flieller et al. \(2006\)](#).

For the special case $A_{q_i} = A_c$ and $\mathbf{b}_{q_i} = B_c \mathbf{u}_{i-1}$,

$$\Phi = e^{A_c T}, \quad \Gamma_{\tau,i} = e^{A_c (T - \bar{t}_i)} B_c (\mathbf{u}_{i-1} - \mathbf{u}_i), \quad (15)$$

which is the fixed-period switched-actuator model of [Marcolino et al. \(2021\)](#), with the corresponding

input-jump sign convention. Mode-dependent state matrices require the full expression in [\(14\)](#).

2.4 Raw proportional timing feedback

A fixed gain $K \in \mathbb{R}^{(N-1) \times n}$ requests

$$\delta \tau_k^{\text{raw}} = -K \mathbf{e}_k, \quad A_{cl} = \Phi - \Gamma_{\tau} K. \quad (16)$$

The gain can be designed from [\(12\)](#), for example by discrete LQR. Here the gain is fixed before conditioning and certification. Although a Schur A_{cl} gives a locally stabilizing raw law, it does not make every requested schedule admissible. In [figure 1\(c\)](#), the raw request moves the last interior instant beyond T , making $d_N = T - t_{k,N-1} < 0$. A minimum dwell constraint is stricter still: it requires $t_{k,N-1} \leq T - d_{\min}$ even before the boundaries

cross. The next section enforces all such adjacent-dwell constraints; [section 4](#) examines the resulting conditioned dynamics.

3 Solver-free dwell-time conditioning

Minimum dwell durations couple adjacent offsets through D . Independent clipping need not preserve the direction of the feedback action. Instead, we apply one scalar factor to the complete raw vector.

Assume that every nominal interval has strict margin,

$$\mathbf{m} = \bar{\mathbf{d}} - d_{\min} \mathbf{1} > \mathbf{0}. \quad (17)$$

The Raw-Action Admissible Region is

$$\mathcal{C} = \{\mathbf{e} : D\mathbf{K}\mathbf{e} \leq \mathbf{m}\}. \quad (18)$$

Inside $\mathcal{C}$, the raw action already satisfies all dwell constraints. Outside it, at least one requested interval is too short. With $\mathbf{q}_k = D\delta\boldsymbol{\tau}_k^{\text{raw}}$, define

$$\beta_k = \min \left\{ 1, \min_{i: q_{k,i} < 0} \frac{m_i}{-q_{k,i}} \right\}, \quad (19)$$

$$\delta\boldsymbol{\tau}_k^{\text{app}} = \beta_k \delta\boldsymbol{\tau}_k^{\text{raw}},$$

where a minimum over an empty set is $+\infty$. Only intervals that the raw action would shorten impose upper bounds on β_k .

Proposition 1 (Feasibility and maximality). *Under (17), (19) returns the largest $\beta \in [0, 1]$ for which every applied dwell satisfies $\bar{\mathbf{d}} + \beta D\delta\boldsymbol{\tau}_k^{\text{raw}} \geq d_{\min} \mathbf{1}$. For a finite raw action, $\beta > 0$. The conditioner preserves the direction of a nonzero raw action, mode order, and cycle period. Computing the factor takes $O(N)$ time without an online solver.*

Proof For $q_{k,i} \geq 0$, $m_i + \beta q_{k,i} > 0$ for all $\beta \in [0, 1]$. For $q_{k,i} < 0$, feasibility is equivalent to $\beta \leq m_i / (-q_{k,i})$. Intersecting these bounds with $[0, 1]$ gives (19), proving feasibility and maximality. Every finite bound is positive. Uniform positive scaling preserves direction, and $\mathbf{1}^\top D = 0$ preserves T . Since $d_{\min} > 0$, the boundaries remain ordered. Adjacent differences compute $\mathbf{q}_k$, and one scan evaluates the N bounds. $\square$

At each cycle start, the controller therefore measures $\mathbf{x}_k$, computes the raw action, scans the dwell

changes, and applies the scaled offsets. If $\beta_k < 1$, at least one dwell reaches its minimum. No set-membership test or optimization is required online. The polyhedron $\mathcal{C}$ explains where conditioning is inactive. [Section A](#) compares the feasible regions for smaller fixed factors and checks their invariance under the corresponding linearized dynamics.

4 Lyapunov stability and convex optimization

Feasibility alone does not imply convergence. This section gives a sufficient stability condition for the complete conditioned linearized family, including cycles in which $\beta_k < 1$.

4.1 Conditioned matrix family

Let S_x be a fixed positive diagonal state-scale matrix and set $\mathbf{z}_k = S_x^{-1} \mathbf{e}_k$. Define the normalized endpoints

$$A_0 = S_x^{-1} \Phi S_x, \quad A_1 = S_x^{-1} A_{\text{cl}} S_x. \quad (20)$$

Substituting (19) into the first-order model gives

$$\mathbf{z}_{k+1} = A(\beta_k) \mathbf{z}_k, \quad (21)$$

$$A(\beta) = (1 - \beta)A_0 + \beta A_1.$$

Equation (21) omits the nonlinear remainder in (12). Although the implemented conditioner returns $\beta > 0$ for finite errors, certifying the closed interval $[0, 1]$ includes its limiting zero-feedback endpoint. Similarity by S_x changes neither endpoint's eigenvalues.

Schur stability of A_0 and A_1 separately is insufficient to establish stability of every matrix in the segment or of their time-varying products. We seek one quadratic function that decreases for both endpoints.

4.2 Common quadratic Lyapunov certificate

Theorem 1 (Uniform stability under conditioning). *Suppose there exist $P = P^\top \succ 0$ and $\eta > 0$ such that*

$$P - A_i^\top P A_i \succeq \eta I_n, \quad i \in \{0, 1\}. \quad (22)$$

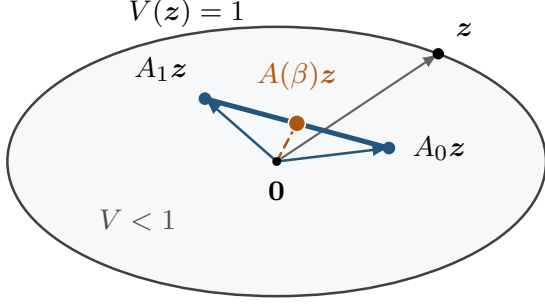


Figure 2 Schematic common- P geometry in normalized error coordinates. The ellipse is the unit level set of V ; the segment joins the endpoint images $A_0 z$ and $A_1 z$, with an intermediate image $A(\beta)z$ marked.

Then $V(z) = z^\top P z$ decreases for every nonzero state of (21), uniformly over $\beta \in [0, 1]$. For any sequence $\{\beta_k\} \subset [0, 1]$,

$$\begin{aligned} \|z_k\|_2 &\leq \sqrt{\frac{\lambda_{\max}(P)}{\lambda_{\min}(P)}} r^k \|z_0\|_2, \\ r &= \sqrt{1 - \frac{\eta}{\lambda_{\max}(P)}} < 1. \end{aligned} \quad (23)$$

In particular, the result holds for the state-dependent factor in (19).

Proof For the same $P \succ 0$ and $\eta > 0$ at both endpoints, define

$$\mathcal{S}_\eta(\beta) = \begin{bmatrix} P - \eta I_n & A(\beta)^\top P \\ P A(\beta) & P \end{bmatrix}. \quad (24)$$

Since $P \succ 0$, the Schur complement of its lower-right block gives the equivalence

$$\mathcal{S}_\eta(\beta) \succeq 0 \iff P - A(\beta)^\top P A(\beta) \succeq \eta I_n.$$

The endpoint conditions therefore give $\mathcal{S}_\eta(0), \mathcal{S}_\eta(1) \succeq 0$. The affine dependence on β yields

$$\begin{aligned} \mathcal{S}_\eta(\beta) &= (1 - \beta)\mathcal{S}_\eta(0) \\ &\quad + \beta\mathcal{S}_\eta(1) \succeq 0 \end{aligned} \quad (25)$$

for every $\beta \in [0, 1]$, by convexity of the positive semidefinite cone. The Schur complement then gives, for $z_{k+1} = A(\beta_k)z_k$,

$$\begin{aligned} V(z_{k+1}) - V(z_k) &\leq -\eta \|z_k\|_2^2 \\ &\leq -\frac{\eta}{\lambda_{\max}(P)} V(z_k). \end{aligned} \quad (26)$$

The same bound holds independently at every cycle, without any assumption on how β_k is selected. Iteration and the inequalities $\lambda_{\min}(P)\|z\|_2^2 \leq V(z) \leq \lambda_{\max}(P)\|z\|_2^2$ give (23). $\square$

Figure 2 illustrates why the same P matters. For every z with $V(z) = 1$, both endpoint images lie strictly inside the same ellipsoid. Their convex combination $A(\beta)z$ lies on the segment joining them, which remains inside that convex sublevel set. Using one V at every cycle makes the decrease comparable even when β_k changes. The drawing is a two-dimensional schematic, not a converter trajectory or a nonlinear region of attraction.

The Schur-complement argument proves decrease on the entire matrix segment, not only at sampled values of β . It certifies the linearized conditioned model for all initial errors in that model, including those outside $\mathcal{C}$. It does not make the linear approximation valid at arbitrary physical errors.

4.3 Offline semidefinite program

For a fixed gain and fixed endpoint matrices, the certificate can be found by

$$\begin{aligned} &\underset{P=P^\top, \eta}{\text{maximize}} && \eta \\ &\text{subject to} && \text{tr}(P) = n, \\ & && P \succeq \varepsilon_P I_n, \\ & && P - A_0^\top P A_0 \succeq \eta I_n, \\ & && P - A_1^\top P A_1 \succeq \eta I_n, \end{aligned} \quad (27)$$

with $\varepsilon_P = 10^{-7}$. The trace constraint removes the arbitrary positive scale of P . For fixed A_0, A_1 , both these inequalities and their equivalent inverse-free block form $\mathcal{S}_\eta(0), \mathcal{S}_\eta(1) \succeq 0$ are affine in P, η . The Julia implementation uses the smaller inequalities in (27). The gain K is fixed, not jointly optimized. A stability certificate requires a positive verified decrease margin, not merely successful solver termination.

After solving, we symmetrize the returned P and recompute

$$\begin{aligned} Q_i &= P - A_i^\top P A_i, \quad i \in \{0, 1\}, \\ \mu &= \min\{\lambda_{\min}(Q_0), \lambda_{\min}(Q_1)\}. \end{aligned} \quad (28)$$

Positive definiteness, trace normalization, and $\mu > 0$ are checked directly. The verified margin μ can replace η in (23). Maximizing η under trace normalization does not directly minimize the resulting rate bound, which also depends on $\lambda_{\max}(P)$.

The certificate is sufficient, not necessary. In particular, including $\beta = 0$ requires A_0 to be Schur. An unstable open-loop endpoint cannot satisfy this full-interval test, even if a more restricted state-dependent controller might stabilize it. Failure to obtain a positive common margin does not by itself establish instability.

4.4 What follows for the exact plant

The exact conditioned cycle map retains a state-dependent timing remainder. Since $\|\delta\tau^{\text{app}}\| \leq \|K\|\|e\|$, its normalized local form is

$$z_{k+1} = A(\beta_k)z_k + \mathcal{O}(\|z_k\|^2). \quad (29)$$

The common- P proof does not bound that remainder over a large-error region.

There is nevertheless a local exact result. Strict nominal dwell margin places a neighborhood of $e = \mathbf{0}$ inside $\mathcal{C}$. There the conditioner is inactive, and the exact closed-loop map is smooth with Jacobian A_{cl} . The endpoint inequalities imply that this Jacobian is Schur. The discrete-time linearization theorem therefore gives local exponential stability of the cycle anchor, in a sufficiently small invariant neighborhood. Smooth interval flows and switching offsets tending to zero extend convergence to the phase-aligned reference periodic trajectory between cycle starts. This argument does not claim nonlinear invariance of the entire polyhedron $\mathcal{C}$.

Open loop is a distinct case. With fixed nominal durations, (13) gives $\mathcal{P}(x, \mathbf{0}) = \Phi x + h$ exactly. Subtracting its fixed point yields

$$e_{k+1} = \Phi e_k \quad (30)$$

without a linearization remainder. Thus a Schur Φ proves convergence under the prescribed open-loop schedule throughout this ideal affine model. This fact does not extend to the nonlinear timing-feedback map.

5 Application to a three-cell converter

The application checks the cycle model, evaluates the common- P certificate, and compares exact controlled and open-loop trajectories. Since the

benchmark is already open-loop stable, the practical question is how much feedback accelerates convergence while preserving dwell feasibility.

5.1 Benchmark and controller parameters

We use the three-cell multilevel DC–DC converter benchmark of Patino et al. [Patino et al. \(2010\)](#), with $x = [v_{C_1}, v_{C_2}, i_L]^T$. For binary switch commands $u_1, u_2, u_3 \in \{0, 1\}$,

$$\begin{aligned} A_u &= \begin{bmatrix} 0 & 0 & (u_2 - u_1)/C_1 \\ 0 & 0 & (u_3 - u_2)/C_2 \\ (u_1 - u_2)/L & (u_2 - u_3)/L & -R/L \end{bmatrix}, \\ b_u &= \begin{bmatrix} 0 \\ 0 \\ Eu_3/L \end{bmatrix}. \end{aligned} \quad (31)$$

Both matrices depend on the switch state, so the common-state-matrix reduction in (15) does not apply. The fixed nine-interval physical mode sequence is $(0, 1, 3, 7, 2, 0, 4, 7, 4)$, where the physical mode ID is $4u_1 + 2u_2 + u_3$.

The published cycle boundaries are $(0, 66, 88, 110, 132, 154, 220, 242, 264, 286)$ μs , with anchor $x^* = (9.9247, 19.2928, 0.9823)^T$. These rounded data do not close exactly. A one-time least-squares dwell reconciliation preserves the reported anchor, 286 μs period, and 22 μs nominal-design bound. It gives

$$\bar{t} = (0, 63.889, 87.557, 109.557, 132.538, 154.538, 218.210, 240.380, 263.828, 286) \mu\text{s}. \quad (32)$$

Computations use the full-precision schedule, with closure error $1.421\text{e-}14$ in the infinity norm and maximum boundary correction 2.111 μs . This reconciliation prepares the benchmark; it is not a proposed reference-trajectory design method.

[Figure 1\(a\)](#) shows the reconciled schedule. The corresponding reference periodic trajectory and cycle anchor are overlaid in [figures 3](#) and [5](#). Its cycle averages are 10.410 56 V, 19.422 28 V, and 0.956 13 A; these averages are not the feedback target.

With $v = \delta\tau/t_s$, the normalized timing-input matrix is $B_n = S_x^{-1}\Gamma_\tau t_s$. Discrete LQR on (A_0, B_n) yields $v^{\text{raw}} = -K_n z$, or $K = t_s K_n S_x^{-1}$ in physical coordinates. [Table 1](#) lists the weights.

Table 1 Converter and timing-feedback parameters. The applied dwell bound is a simulation assumption, not a hardware-calibrated limit.

Quantity	Value
Source E	30 V
Capacitances C_1, C_2	40 μ F
Inductance L ; resistance R	10 mH; 10 Ω
Intervals N ; period T	9; 286 μ s
Nominal-design dwell bound	22 μ s
Applied dwell bound $d_{\min}$	3 μ s
State scale S_x	diag(10, 20, 1)
Timing scale t_s	10 μ s
LQR state weight Q_n	I_3
Aggressive input weight R_n	0.001 I_8
Conservative input weight R_n	0.1 I_8

Only the aggressive law is conditioned; the conservative law is an unconditioned comparison that remains dwell-feasible for this experiment.

5.2 Model checks and numerical Lyapunov certificate

Central finite differences of the exact map check both Jacobians. State steps are 10^{-4} times the diagonal entries of S_x , and the timing step is $10^{-3}t_s = 0.01 \mu$ s. The relative errors for Φ and Γ_τ are 8.518e-12 and 8.344e-11, respectively. For simultaneous normalized state and timing perturbations of amplitude ϵ , we also evaluate

$$r_{\text{lin}}(\epsilon) = \|S_x^{-1}[\mathcal{P}(\mathbf{x}^* + \Delta\mathbf{x}, \delta\tau) - \mathbf{x}^* - \Phi\Delta\mathbf{x} - \Gamma_\tau\delta\tau]\|_2. \quad (33)$$

The fitted log-log slope is 2.000, consistent with the second-order remainder.

For the aggressive gain, the Julia implementation solves (27) using Convex.jl and Clarabel with $\text{tr}(P) = 3$. It returns optimal status and $\eta = 2.911904315 \times 10^{-4}$. The common matrix, displayed to six decimal places, is

$$P \approx \begin{bmatrix} 0.608770 & -0.013731 & 0.027354 \\ -0.013731 & 2.384502 & 0.033889 \\ 0.027354 & 0.033889 & 0.006728 \end{bmatrix}. \quad (34)$$

Full-precision eigenvalue checks give

$$\begin{aligned} \lambda_{\min}(P) &= 4.987946970 \times 10^{-3}, \\ \lambda_{\min}(Q_0) &= 2.911904317 \times 10^{-4}, \\ \lambda_{\min}(Q_1) &= 2.911956096 \times 10^{-4}. \end{aligned} \quad (35)$$

Using the verified $\mu = 2.911904317 \times 10^{-4}$ in (23) gives $r = 0.999938954110$. This worst-case bound accommodates arbitrarily weak correction and is correspondingly slow. It is not a prediction of the aggressive controller's observed settling time.

The Julia solve uses endpoint matrices from a Python reconstruction of the physical model, rather than independently reconstructing the converter. The article pipeline compares those endpoints against its MATLAB model and aggressive gain, then recomputes the inequalities using that model and the returned P . The machine-readable certificate retains full precision; the displayed rounded matrix is checked separately. These are floating-point numerical checks, not interval-arithmetic certification.

5.3 Controlled and open-loop trajectories

All comparison runs start from

$$\mathbf{x}_0 = [7.5143 \quad 20.8211 \quad 0.0314]^T, \quad (36)$$

with normalized error 0.983946. Exact matrix exponentials propagate each actual interval. The simulations do not replace the plant by its linearization. For the continuous-time plots, each interval is sampled at 30 equally spaced local times, including its end, rather than drawing straight lines only between switching states.

Figure 3(a) compares the physical trajectories over 100 cycles. The controlled trajectory rapidly approaches the reference. Open loop remains visibly displaced over this short horizon, but the longer error history in panel (b) shows its eventual approach. Both runs target the same periodic trajectory; feedback accelerates convergence rather than stabilizing an otherwise unstable open-loop benchmark. The dot and star in panel (a) mark the initial state and reference cycle anchor, respectively.

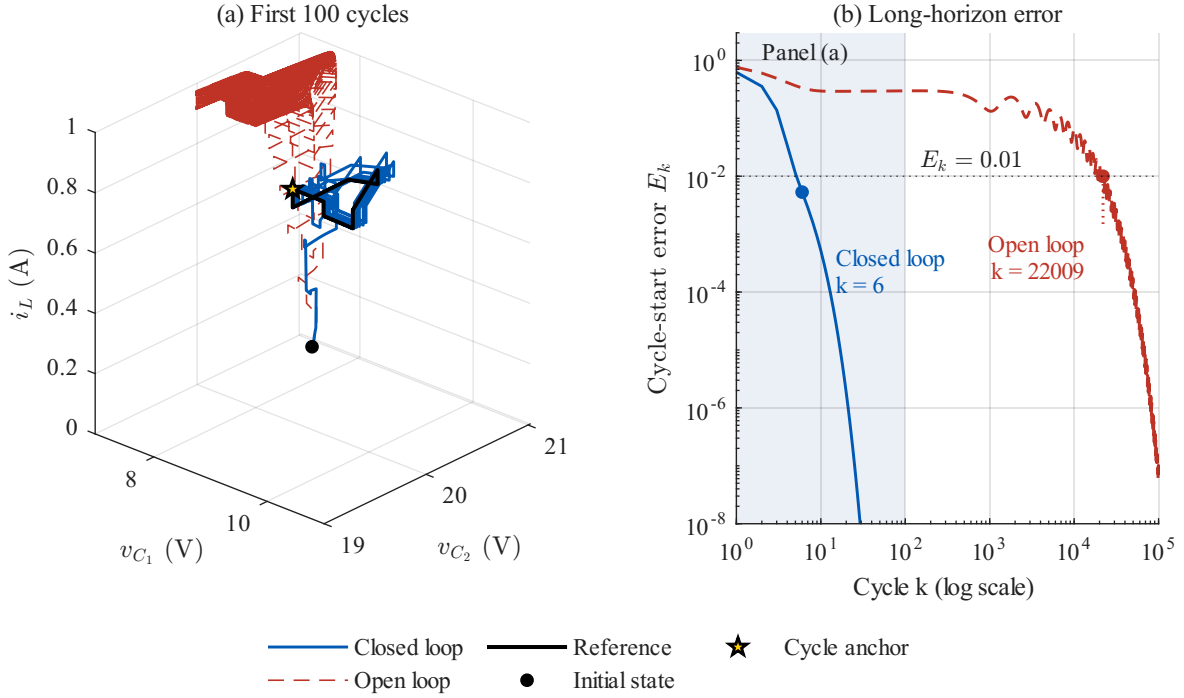


Figure 3 Early transient and eventual convergence. (a) Continuous-time state trajectories over 100 cycles. (b) Normalized cycle-start error on logarithmic axes. Shading marks the horizon in (a); dots mark the saved threshold-crossing samples.

We report the dimensionless normalized cycle-start error

$$E_k = \|S_x^{-1}(\mathbf{x}(kT) - \mathbf{x}^*)\|_2, \quad (37)$$

where $\mathbf{x}^*$ is the cycle anchor and $S_x = \text{diag}(10, 20, 1)$ uses voltage scales of 10 V and 20 V and a current scale of 1 A. Sampling at the same phase distinguishes tracking error from the reference's switching ripple. Under the fixed nominal schedule, convergence of these samples to the anchor also implies convergence of the within-cycle motion to the reference periodic trajectory, not to a constant physical state.

In figure 3(b), cycle zero is omitted because the horizontal axis is logarithmic. Feedback data extend to cycle 100, open-loop data to cycle 100000, and errors below 10^{-8} lie outside the displayed range. The marked crossings are actual samples, not interpolated intersections: the first feedback sample below 0.01 and the open-loop sample after which every remaining simulated error stays below it.

The value $E_k = 0.01$ is a selected reporting threshold for comparing convergence times, not a stability boundary or a hardware accuracy requirement. It is relative to the chosen state scales, not to the initial error or each reference component. The condition $E_k < 0.01$ implies cycle-start component errors below 0.10 V, 0.20 V, and 0.01 A, respectively; the joint Euclidean-norm condition is stricter than these individual bounds alone. Crossing the threshold does not mean that the error is zero. The continued error decay in figure 3(b), together with the exact open-loop error map, establishes the distinction between a reported threshold time and eventual convergence. Figure 4 details the early error and conditioning histories.

Figure 4 aligns the error with the fraction of each timing request that is applied. In this simulation, the first three requests require factors of approximately 0.294, 0.447, and 0.779, respectively. From cycle 3 onward, the complete request is feasible and $\beta_k = 1$. Feedback remains active, but its commands no longer require scaling. The cycle-start error continues to decrease and first falls

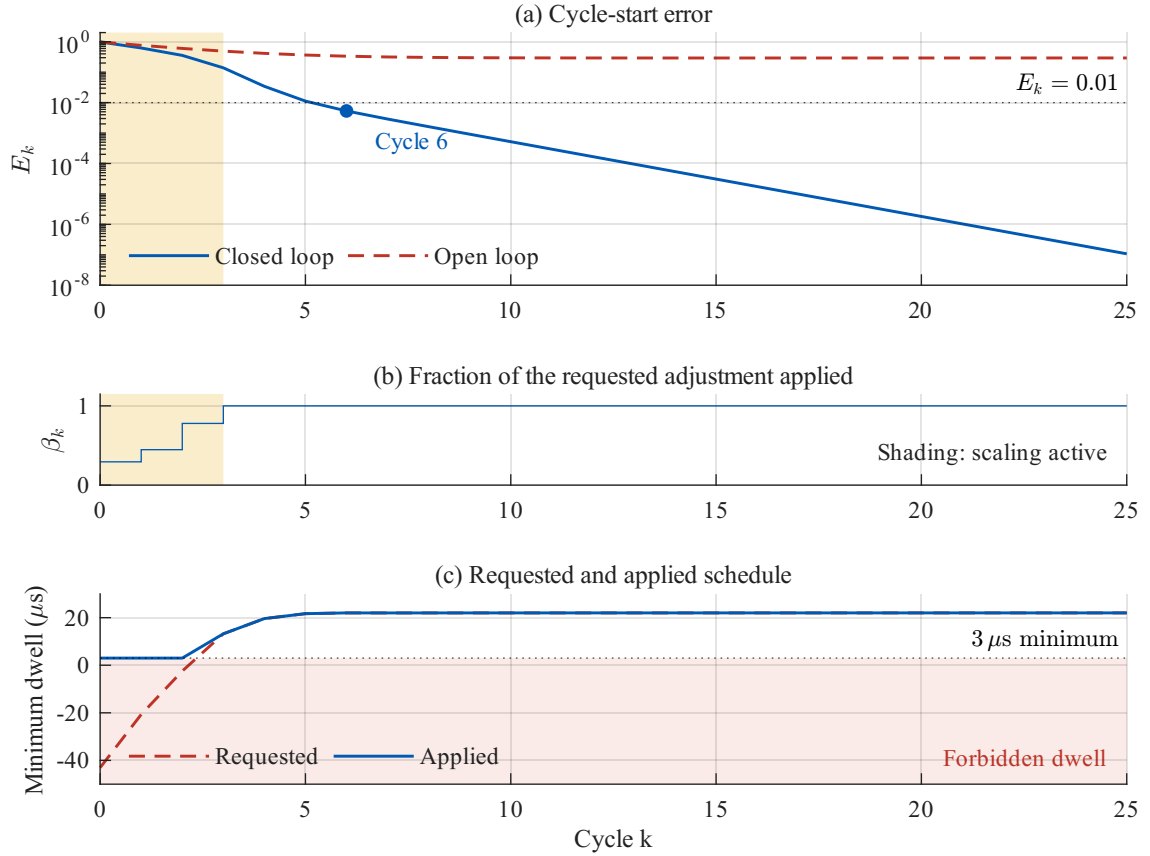


Figure 4 First 25 cycles of the exact simulation. (a) Cycle-start error. (b) Applied fraction of the timing request. Shading in (a)–(b) marks active scaling. (c) Requested and applied minimum dwell, with the forbidden range shaded.

below the selected 0.01 threshold at cycle 6. Scaling is transient in this run; this observation does not establish that it becomes permanently inactive for other initial conditions or after disturbances.

The raw actions in figure 4(c) are candidate commands only; the plant uses the conditioned schedule. The aggressive raw law requests a minimum dwell of $-43.045 \mu s$, which would invert an interval. Conditioning acts during 3 cycles, with minimum factor 0.293978. It reduces the largest absolute offset from 65.217 to $19.172 \mu s$ and keeps every applied dwell at or above $3 \mu s$. The active constraint reaches this bound, as the maximal-scaling formula predicts.

The conservative law remains feasible without conditioning but crosses the error threshold at cycle 20, compared with cycle 6 for the conditioned aggressive law. Table 2 reports the comparison. These large-error nonlinear trajectories support

the controller’s performance in this benchmark; the common- P theorem alone does not certify their entire transient.

5.4 Physical states and long-horizon comparison

Figure 5 separates the physical states along a common time axis, making the individual voltage and current transients visible. Both responses start from the same initial state. The reference repeats the nominal within-cycle waveform, not its average or a constant anchor value. The closed-loop states approach this periodic waveform while open loop remains displaced over the displayed window. The continuing switching ripple is part of the reference motion, not residual cycle-start tracking error. These curves use the exact within-interval samples, rather than only the states at cycle starts.

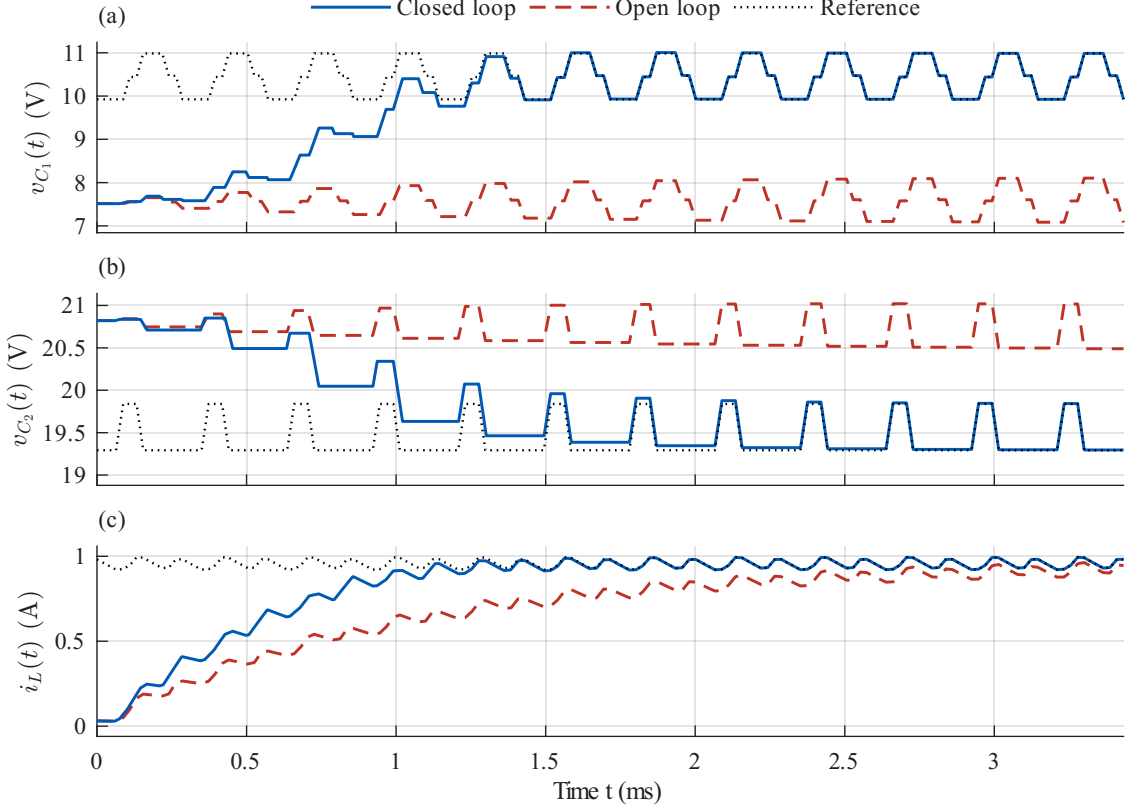


Figure 5 Continuous-time state responses over the first 12 cycles. (a) $v_{C1}(t)$. (b) $v_{C2}(t)$. (c) $i_L(t)$. Each panel compares closed loop, open loop, and the reference periodic waveform.

For the same nominal schedule, $\rho(\Phi) = 0.999850 < 1$. Equation (30) therefore predicts eventual open-loop convergence, not merely local convergence of a linear approximation. The separate 100000-cycle exact simulation in figure 3(b) tests the long time scale hidden by the short comparison. Its normalized cycle-start error remains below 0.01 from cycle 22009 through the end of the run. Since $T = 286 \mu\text{s}$, the open-loop transient spans seconds, whereas the controlled threshold crossing occurs within a few milliseconds.

6 Discussion

Uniform conditioning connects two properties that must be distinguished. The dwell inequalities give a closed-form maximal feasible scale for any finite raw action. The common quadratic Lyapunov inequalities then give a sufficient convergence certificate for the associated linearized

matrix segment. Both the SDP and the supporting invariant-region computation are offline. Online execution remains a static gain multiplication followed by adjacent differences and a minimum scan.

The one-cycle sensitivity follows Patino et al. [Patino et al. \(2010\)](#), and the common-state-matrix reduction recovers Marcolino et al.’s fixed-period timing model [Marcolino et al. \(2021\)](#). Direction-preserving scaling is also established in control allocation [Nakamura-Zimmerer and Miller \(2026\)](#). The contribution here is the closed-form specialization to coupled dwell inequalities and its connection to a common- P certificate for conditioned timing feedback, not a new LQR or a new general Lyapunov theorem.

The full-interval certificate is conservative. It must include the slow open-loop endpoint and use the same quadratic function everywhere. It neither proves that every stabilizing timing gain admits

Table 2 Exact simulation results. Threshold cycles for the two feedback laws denote the first sample with $\|S_x^{-1}e_k\|_2 < 0.01$. The open-loop value denotes the crossing after which every remaining simulated sample stays below 0.01.

Quantity	Value
$\rho(\Phi)$, open loop	0.999850
$\rho(A_{cl})$, aggressive	0.568151
Spectral radius, conservative	0.941468
Minimum β ; conditioned cycles	0.293978; 3
Maximum raw offset	65.217 μ s
Maximum applied offset	19.172 μ s
Minimum raw dwell	-43.045 μ s
Minimum applied dwell	3.000 μ s
Minimum conservative dwell	15.502 μ s
Threshold cycle, conditioned	6
Threshold cycle, conservative	20
Persistent crossing, open loop	22009

such a certificate nor bounds the nonlinear remainder at arbitrary errors. For the benchmark, open loop already approaches the reference, so the benefit of feedback is faster convergence with feasible schedules. Claims of global nonlinear stabilization by the conditioned controller would go beyond these results.

The assumed 3 μ s bound has not been calibrated to hardware. Timer quantization, delays, dead time, implementation margin, state estimation, parameter uncertainty, and experiments remain outside the study. The mode sequence, period, and reference phase are fixed throughout.

7 Conclusion

The proposed conditioner applies the largest dwell-feasible scalar multiple of a proportional timing action in $O(N)$ time, preserving its direction and the fixed cycle period. A common quadratic Lyapunov function found through an offline end-point SDP certifies uniform exponential stability of the conditioned linearized model for every conditioning-factor sequence in $[0, 1]$. Strict nominal dwell margin also yields a local exact nonlinear stability result. In the converter application, exact 3D trajectories and error histories show that conditioning makes the aggressive action feasible and accelerates convergence relative to an already stable open-loop schedule. The long-horizon simulation confirms convergence to periodic motion.

Extending the feedback guarantee beyond a local nonlinear neighborhood requires an explicit remainder or nonlinear Lyapunov analysis.

Appendix A Feasible-region expansion under timing-action scaling

A.1 Expansion of the dwell-feasible region

Reducing the fraction of a timing command makes that command feasible for larger cycle-start errors. For a fixed factor $\beta \in (0, 1]$, the action $\delta\tau = -\beta K e$ satisfies the dwell bounds on

$$\mathcal{F}_\beta = \{e : \beta D K e \leq m\} = \frac{1}{\beta} \mathcal{C}, \quad (\text{A1})$$

$$\mathcal{F}_1 = \mathcal{C}.$$

Thus every radial extent grows by $1/\beta$ relative to the raw-action admissible region. For $0 < \beta_a < \beta_b \leq 1$, $\mathcal{F}_{\beta_b} \subseteq \mathcal{F}_{\beta_a}$. In the three-state benchmark, the volume grows by β^{-3} . The factors 0.25, 0.5, and 1 therefore give volume ratios 64, 8, and 1, respectively. This scaling follows from the dwell inequalities; by itself it is a current-cycle feasibility result, not a stability claim.

Figure A1 compares these nested regions. The trajectory uses the maximal state-dependent factor, not one of the fixed comparison factors throughout. Its initial error lies inside $\mathcal{F}_{0.25}$ but outside $\mathcal{F}_{0.5}$, consistent with the first applied factor of approximately 0.294. The raw action becomes feasible at cycle 3. Lines join cycle-start samples, not within-cycle trajectories. The planar views are projections, not zero-error slices; membership and entry are evaluated using the full three-dimensional inequalities.

A.2 Feasibility over successive cycles

Current-cycle feasibility does not alone guarantee that the next error remains in the same feasible region. Holding a comparison factor β fixed gives the linearized physical-error dynamics

$$e_{k+1} = M_\beta e_k, \quad M_\beta = \Phi - \beta \Gamma_\tau K. \quad (\text{A2})$$

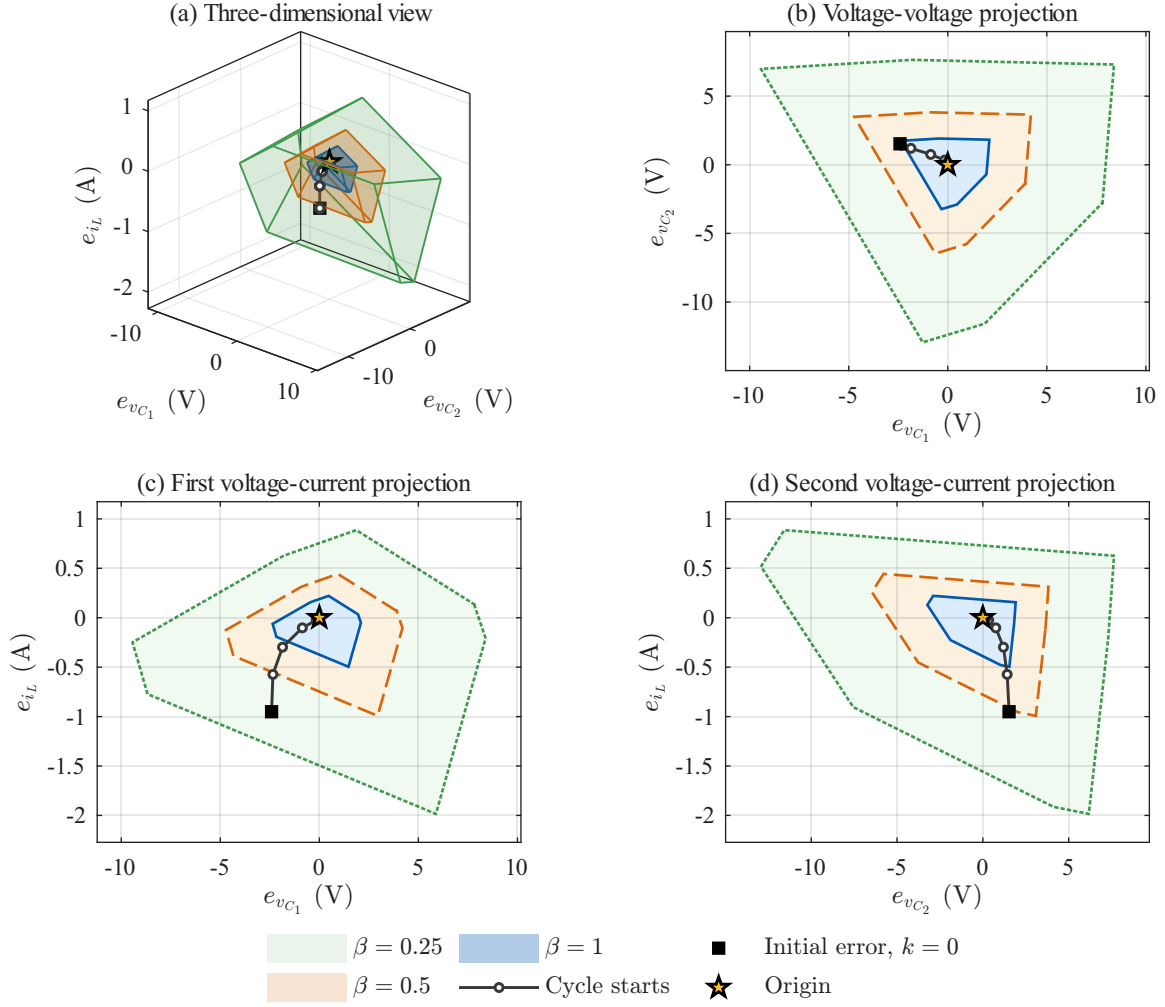


Figure A1 Dwell-feasible regions for fixed factors $\beta = 0.25, 0.5, 1$, with early cycle-start errors from the conditioned simulation. (a) Three-dimensional view. (b)–(d) Coordinate-plane projections. Smaller factors admit larger errors.

For each of the three plotted factors, we checked the positive-invariance condition

$$M_\beta \mathcal{F}_\beta \subseteq \mathcal{F}_\beta. \quad (\text{A3})$$

The offline polyhedral calculation uses MPT3 [Herceg et al. \(2013\)](#) and the preimage-containment approach of [Gilbert and Tan \(1991\)](#). All three regions pass for this benchmark. Equivalently, each vertex $\mathbf{v}$ satisfies $\beta DKM_\beta \mathbf{v} \leq \mathbf{m}$ within numerical tolerance. Linearity and convexity extend the vertex checks to every point in the polyhedron, so

repeated application of the same fixed-factor controller keeps its linearized trajectory dwell-feasible.

Since $S_x^{-1} M_\beta S_x = A(\beta)$, [theorem 1](#) also guarantees convergence of each fixed-factor linearized model. Thus every initial error in a plotted region remains feasible and converges under its corresponding fixed factor. The case $\beta = 1$ is simply the raw controller on $\mathcal{F}_1 = \mathcal{C}$; the smaller factors admit larger regions of feasible linearized operation.

These region checks concern fixed factors, whereas the simulated controller selects β_k from the current error. They do not establish nonlinear invariance of the plotted polyhedra or an enlarged

region of attraction for the exact plant. Exact nonlinear stability remains local, and the displayed large-error trajectory is simulation evidence. The common- P theorem covers the conditioned linearized family independently of these polyhedral checks.

Declarations

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Data availability. TODO: state where the generated simulation data are available, or that all data supporting the findings are contained in the article. Required for original research.

Code availability. TODO: state where the simulation and figure-generation scripts are available, or that this item is not applicable.

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